

FORENSIC INCIDENT ANALYSIS REPORT

Threat Report: njRAT Remote Access Trojan & Backdoor Intrusion

SEVERITY: CRITICAL

PREPARED BY	Dhruva Khare	TARGET HOST	SERVER2022
TARGET IP	10.10.1.22	DOMAIN SUFFIX	CEH.com
OPERATING SYSTEM	Windows Server 2022 Std x64	USER CONTEXT	Administrator

1. EXECUTIVE SUMMARY

A comprehensive digital forensics and incident response (DFIR) investigation was conducted on target host **SERVER2022** (IP: **10.10.1.22**). The analysis confirms an active, high-severity system compromise involving the **njRAT v0.8d** Remote Access Trojan (RAT) alongside an auxiliary backdoor binary, **tini.exe**.

The threat actor established interactive Remote Access Control, actively surveilled local user sessions, established direct Command and Control (C2) communications, and successfully exfiltrated unencrypted sensitive financial credentials from the administrative session.

2. SYSTEM PROFILE & TARGET ENVIRONMENT

CONFIGURATION FIELD	SYSTEM VALUE / INDICATOR
Host Infrastructure	Server2022.CEH.com
Network Configuration	IPv4: 10.10.1.22 Mask: 255.255.255.0 Gateway: 10.10.1.2
Network Adapter	Microsoft Hyper-V Network Adapter (MAC: 00-15-5D-01-80-02)
OS Architecture	Windows Server 2022 Standard SP0 x64
Session Privilege	Administrator (Elevated Full Access)

3. ADVERSARY ACTIVITY & THREAT CAPABILITIES

A. njRAT C2 Interaction Capabilities

- Process Controls:** Direct administrative manipulation observed (**Kill** , **Suspend** , **Kill + Delete** execution).
- Registry Modification:** Unrestricted hive access and key alteration under **HKEY_CLASSES_ROOT**.
- Desktop Surveillance:** Real-time screen capture and desktop session monitoring active via C2 port **5553**.
- Interactive Session Hijacking:** Active console messaging initiated by attacker (**Chating With >> Hacker**).

B. Credential Compromise & Exfiltration Event

CRITICAL SECURITY BREACH — DATA EXFILTRATION CONFIRMED

Unencrypted financial credentials were stolen during an active Notepad editing session:

Exfiltrated String: "My credit card number is 7863654646575"

Adversary C2 Message Transmitted: [Hacker] hello mr what you are doing

4. STATIC & TECHNICAL MALWARE ANALYSIS

Sample 1: Backdoor Payload (tini.exe)

ARTIFACT ATTRIBUTE	FORENSIC INDICATOR
Filename & Format	tini.exe (PE32 executable / application/x-dosexec)
File Size	3 KiB (3,072 bytes)
SHA-256 Hash	9654bb748199882b0fb29b1fa597c0cfe3b9d610adf4188a0b440f3faf5ee527
Threat Assessment	100/100 (Malicious) — 93% Detection Rate (Hybrid Analysis)
MetaDefender Result	23 / 27 Engines Flagged
AV Signatures	Backdoor.Win32.Generic , Win-Trojan/IQ.B , Backdoor.Win32.Tiny.B , Troj/Cmdoor-A

Sample 2: Binary Inspection via Detect It Easy (DiE)

- **Format:** ELF64 (X86-64 Architecture)
- **Entropy Analysis:** Total Entropy = 2.87901 (Status: **Unpacked - 35%**). Low entropy confirms raw executable binary code suitable for reverse engineering.
- **Key Disassembly Opcodes:**

```
0000:0490 31ed      xor ebp, ebp
0000:0492 4989d1    mov r9, rdx
0000:0495 5e        pop rsi
0000:0496 4889e2    mov rdx, rsp
0000:0499 4883e4f0  and rsp, 0xfffffffffffffff0
0000:049d 50        push rax
0000:049e 54        push rsp
0000:04b4 e8a7ffff  call 0x460
```

5. NETWORK TELEMETRY ANALYSIS (TCPVIEW OUTPUT)

PROCESS	PID	PROTO	STATE	LOCAL ENDPOINT	REMOTE ENDPOINT
dns.exe	3564	TCP	Listen	10.10.1.22:53	0.0.0.0:0
svchost.exe	520	TCP	Listen	0.0.0.0:135 / 593	0.0.0.0:0
System	4	TCP	Listen	10.10.1.22:139	0.0.0.0:0
lsass.exe	820	TCP	Listen	0.0.0.0:389 / 636	0.0.0.0:0

PROCESS	PID	PROTO	STATE	LOCAL ENDPOINT	REMOTE ENDPOINT
postgres.exe	4628	TCP	Listen	127.0.0.1:7337	0.0.0.0:0
System (C2 Link)	4	TCP	ESTABLISHED	10.10.1.22:60718	10.10.1.11:445

6. REMEDIATION & INCIDENT RESPONSE PLAYBOOK

- Host Isolation:** Immediately disconnect 10.10.1.22 from network switches/VLANs to sever active C2 persistence and prevent lateral movement.
- C2 Containment & Blocking:** Terminate all malicious process trees associated with tini.exe and njRAT. Enforce perimeter firewall blocks for target C2 IP 10.10.1.11.
- Credential Revocation:** Execute an immediate password reset for the domain Administrator account and all active service accounts on SERVER2022.
- Financial Loss Prevention:** Immediately flag exfiltrated credit card number (7863654646575) for fraud monitoring and reissuance with the issuing bank.
- Forensic Imaging:** Acquire volatility memory dumps (RAM) and snapshot registry hives (HKLM , HKCU) prior to host re-imaging.